

Table 1. False-premise detection across models and datasets

Method	Qwen2.5 7B		GPT-6 Luna		Gemma 4 12B		Qwen3.8 27B OFF		Qwen3.8 27B ON	
	FPQ TPR ↑	NFP FPR ↓	FPQ TPR ↑	NFP FPR ↓	FPQ TPR ↑	NFP FPR ↓	FPQ TPR ↑	NFP FPR ↓	FPQ TPR ↑	NFP FPR ↓
Cancer-Myth										
Direct gate	7.0 ^H	10.7 ^H	—	—	40.5 ^H	16.1 ^H	26.1 ^H	16.1 ^H	44.1 ^H	21.5 ^H
Premise-review CoT gate	29.5 ^H	16.1 ^H	—	—	51.3 ^H	32.9 ^H	57.5 ^H	35.6 ^H	61.4 ^H	38.3 ^H
Probe gate (hidden states)	75.0 ^H	47.7 ^H	n/a	n/a	—	—	—	—	—	—
TF-IDF text	Shared detector: FPQ 86.1% / NFP 51.0% (H)									
GEPA	—	—	—	—	—	—	—	—	—	—
Ours	—	—	—	—	—	—	—	—	—	—
CREPE										
Direct gate	—	—	—	—	—	—	—	—	—	—
Premise-review CoT gate	—	—	—	—	—	—	—	—	—	—
Probe gate (hidden states)	—	—	n/a	n/a	—	—	—	—	—	—
TF-IDF text	Shared detector: not measured									
GEPA	—	—	—	—	—	—	—	—	—	—
Ours	—	—	—	—	—	—	—	—	—	—

Stored results; —: not measured/comparable; n/a: outside the defined implementation. Percentages, one run.
H: historical full-corpus Cancer-Myth; valid denominators and protocol differences in ledger.
TPR ↑: FPQ detection; FPR ↓: false positives on normal questions.
Probes require hidden states; no native Luna probe. TF-IDF is model-independent.
Default crossfit probe/TF-IDF decisions; these are not matched-FPR thresholds.
Ours denotes the proposed method; its implementation is not fixed.